

Rasika Karkare, Ph.D.

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scholar.google.com/citations?user=IUFGluwAAAAJ

Professional Summary

Ph.D. Machine Learning Engineer and Data Scientist with 4+ years of experience building production-grade ML, NLP, and generative AI systems across fintech, healthcare, energy, and research domains. Core technical focus: LLM evaluation, retrieval-augmented generation (RAG) pipeline design, parameter-efficient fine-tuning (LoRA/QLoRA), and privacy-preserving analytics, grounded in rigorous statistical experimentation (RCT design, causal inference). Targeting ML/AI Engineer, Senior Data Scientist, or Applied Research Scientist roles centered on trustworthy and responsible AI evaluation.

Professional Experience

Postdoctoral Researcher, ABC Innovation Lab (*incoming*)

Syracuse University School of Information Studies | starting October 16, 2026 | PI: Dr. Lee McKnight

- Will report to Dr. Lee McKnight, in cooperation with the Maxwell School's Dynamic Sustainability Lab, on a multi-year USDA "Advancing Markets for Producers" grant.
- Will build human-centered, agentic AI-enhanced data services and open cloud/edge/blockchain-based tools for producer-facing markets.
- Will mentor undergraduate and graduate student researchers on the project.

Postdoctoral Research Fellow

Worcester Polytechnic Institute, Worcester, MA | August 2025–2026 | PI: Dr. Jacob Whitehill

- Designed a randomized controlled trial (RCT) linking LLM-generated text surprisal (GPT-4, Gemini, LlamaGen) to reader comprehension, reading behavior, and cognitive load, using a fixed open-weights scorer independent of every generator (self-surprisal only measures a model's own decoding temperature, not reader difficulty).
- Built RAG-Eval, a 2×2 factorial framework (retrieval on/off × prompt-optimization on/off), with LoRA fine-tuning as a third comparison arm, scored via RAGAS, Flesch-Kincaid readability, and surprisal metrics; preliminary results show retrieval and prompt engineering act as independent main effects with no significant interaction detected at current sample size.
- Designing an intervention study (in progress) that detects high-surprisal text spans and rewrites them via three arms — zero-shot prompting, retrieval-grounded rewriting, and LoRA fine-tuning — verified by a LangGraph agent loop that checks surprisal reduction, factual entailment, and length before accepting a rewrite, with a bounded retry count and human-review fallback.
- Built a multimodal video anonymization pipeline (Karkare, Cao, Whitehill) using DensePose and a Detectron2 model fine-tuned on 30 hand-annotated classroom images (chairs, desks, whiteboards); ran a randomized experiment (n=40) comparing raw, face-blurred, and rendered video across activity, action, engagement, and gaze-annotation tasks. Found utility is task-dependent, audio removal drives the largest accuracy loss, and anonymization level itself biases perceived student engagement. Manuscript submitted to the Journal of Educational Data Mining, 2026.
- Developed an end-to-end voice anonymization pipeline (Whisper ASR, Parselmouth/Praat prosody extraction, Google Cloud TTS Neural2 with SSML); achieved median pitch match within 1% of original (299.4 Hz vs. 302.9 Hz) and energy match within 1.2 dB across 8 benchmarked voice synthesis systems.

Assistant Director – Data Scientist

Moody's Analytics Corp., New York, NY | August 2022–August 2025

- Built a production multimodal LLM pipeline (Gemini, GPT-4V) for structured information extraction from multilingual PDFs, supporting \$1B+ in analytics workflows, including output validation and quality monitoring.
- Led implementation of LLM-based pipelines extracting sustainability and financial metrics from 100+ multilingual PDF reports, achieving up to a 30% improvement in precision for key metric identification.
- Independently developed and fine-tuned NLP pipelines integrating sentiment analysis and temporal tagging for financial event detection, completing 3–4 related projects per year.

- Enhanced reputation risk modeling by integrating financial news sentiment with quantitative ratios, driving a 15% improvement in reputation risk scoring accuracy against internal and external validation datasets.
- Engineered an entity classification system (Sentence Transformers, LSTM) in Databricks to assign standardized industry codes to under-represented entities, improving data fill rates across large-scale knowledge graph pipelines.
- Developed hierarchical missing-data imputation frameworks using GANs and autoencoders to improve model reliability; designed a stacking-ensemble firm-sizing model (MESA) combining financial, firmographic, and technographic data sources for firmographic prediction.
- Diagnosed model failures for a \$100M+ Graph Convolutional Neural Network (GCNN) initiative on AWS using percentile-based outlier analysis; presented risk assessment and remediation strategy to executive stakeholders.
- Fine-tuned domain-specific BERT models on proprietary text corpora for risk signal detection; built automated data ingestion and model monitoring pipelines in CI/CD environments.
- Co-created and mentored on a GenAI-powered candidate-project matching system for internal team staffing (2nd Place, Moody's Corporation Hackathon): built a full-stack web app in which a recruiter enters an applicant's background and skillset and the backend LLM pipeline matches them against open project/job descriptions, ranking candidates by fit; designed both the matching logic (embedding-based skill/role similarity) and the front-end interface for recruiters to review and act on matches.
- Mentored rotating summer/intern data scientists on NLP and LLM pipeline work, including onboarding to internal data infrastructure and code review.

Machine Learning Research Intern, Predictive Modeling

Philips Healthcare, Burlington, MA | May 2022–August 2022

- Built a sepsis risk prediction model using time-series physiological data (XGBoost, LightGBM), achieving AUC-ROC of 0.91 and reducing false positives by 25% versus rule-based scoring (SIRS, qSOFA) through optimized threshold tuning.
- Developed a depression-detection model from clinical interview audio using STFT and MFCC features with CNNs, achieving F1 of 0.83 on a highly imbalanced dataset; applied SHAP analysis to surface key vocal cues for clinician interpretability.

Quantitative Researcher Intern, Data Science

Moody's Analytics, New York, NY | November 2021–January 2022

- Developed deep learning pipelines (ResNet50, transfer learning) combining quantitative modeling with satellite image preprocessing to assess post-hurricane building damage for insurance risk evaluation, achieving F1 of 0.89 across damage severity categories.

Ph.D. Researcher, Deep Learning and Robust Signal Recovery

Worcester Polytechnic Institute, Worcester, MA | August 2017–August 2022 | Advisors: Dr. Randy Paffenroth, Dr. Diran Apelian

- Developed the Self-Supervised Hadamard Autoencoder framework for blind missing-data imputation, denoising, and inpainting, validated across four domains: image denoising/inpainting (MNIST, CIFAR-10), inferring missing distances in ultra-sparse social networks, estimating intrinsic dimensionality in intraday financial trading data despite missing entries, and missing-sensor-data imputation in manufacturing.
- Integrated robust PCA with self-supervised learning to recover structured signals without clean ground-truth labels, outperforming state-of-the-art matrix completion baselines under extreme missingness.
- Applied the Hadamard autoencoder framework in an industry-sponsored collaboration with the Advanced Casting Research Center (ACRC) at UC Irvine and Mercury Marine, treating missing sensor data in aluminum high-pressure die-casting manufacturing processes.
- Mentored NSF REU (Research Experience for Undergraduates) students on applied deep learning projects.

Data Analyst

Energy Insight, Inc., United States | 2016–2017

- Performed energy audits and analyzed building/facility consumption data to identify efficiency opportunities and support client-facing recommendations.

Education

Ph.D., Data Science, Worcester Polytechnic Institute

M.S., Civil and Environmental Engineering, University of Maryland, Baltimore County
B.E., Engineering, MIT, Pune

Selected Publications

1. Karkare R., Cao G., Whitehill J. *Classroom Video Anonymization: Evaluating the Trade-off Between Video Privacy Preservation and Observation Fidelity*. Journal of Educational Data Mining, 2026. [Under Review]
2. Tlachac M.L., Karkare R. et al. *Video and Image Datasets for Depression Assessment: A Scoping Review*. Transactions on Affective Computing, June 2026. [Under Review]
3. Karkare R. et al. *Blind Image Denoising and Inpainting Using Robust Hadamard Autoencoders*. IEEE Access. [Under Review]
4. Mohandesi G., Karkare R., Paffenroth R., Timko M. *When Accurate Models Mislead: Detecting and Mitigating Overfitting in Waste-to-Energy Machine Learning*. A hierarchical validation method for tree-based regressors across four waste-to-energy datasets, in collaboration with chemical engineering researchers. [In Progress]
5. Medina P., Karkare R. *Integrating Product Coefficients for Improved 3D LiDAR Data Classification (Part II)*. Preprint, October 2025.
6. Karkare R., Paffenroth R., Apelian D. *Self-Supervised Deep Hadamard Autoencoders for Treating Missing Data: A Case Study in Manufacturing*. Integrating Materials and Manufacturing Innovation, March 2022. [Published]
7. Mahindre G., Karkare R., Paffenroth R., Jayasumana A.P. *A Pre-training Oracle for Predicting Distances in Social Networks*. IEEE International Conference on Big Data, 2021, pp. 4126–4135. [Published]
8. Mahindre G., Karkare R., Paffenroth R., Jayasumana A.P. *Inference in Social Networks from Ultra-Sparse Distance Measurements via Pretrained Hadamard Autoencoders*. IEEE LCN, 2020. [Published]
9. Kopper A., Karkare R., Paffenroth R., Apelian D. *Model Selection and Evaluation for Machine Learning: Deep Learning in Materials Processing*. Integrating Materials and Manufacturing Innovation, September 2020. [Published]
10. Sun N., Kopper A., Karkare R., Paffenroth R., Apelian D. *Machine Learning Pathway for Harnessing Knowledge and Data in Material Processing*. International Journal of Metalcasting, September 2020. [Published]

Mentoring, Presentations & Service

- Mentor, NSF REU (Research Experience for Undergraduates) students, Worcester Polytechnic Institute.
- Mentor for rotating summer/intern data scientists, Moody's Analytics.
- Poster presenter, Graduate Research Innovation Exchange (GRIE), Worcester, MA.
- Peer Reviewer, IEEE Access, Artificial Intelligence Review, and other journals.
- Member, Women in Data Science; Member, ACRC Consortium.

Awards

- Academic Excellence Award, Worcester Polytechnic Institute.
- Advanced Casting Research Center (ACRC) Ph.D. Fellowship.
- 2nd Prize, Moody's Corporation Hackathon — GenAI-powered candidate-project matching system.

Technical Skills

Generative AI & LLMs: RAG pipeline design, LLM evaluation (RAGAS, readability/comprehension metrics), hallucination detection, fine-tuning (LoRA, QLoRA), prompt engineering, LangChain, LangGraph, HuggingFace Transformers, multi-agent system design

Machine Learning & NLP: Transformers, BERT, Sentence Transformers, LSTM, Graph Neural Networks, Autoencoders & GANs, XGBoost, self-supervised learning, imbalanced data handling, computer vision (DensePose, Detectron2)

Statistics & Experimentation: RCT design, A/B testing, mixed-effects models, causal inference, uplift modeling, power analysis, Bayesian methods, IRB protocol design, inter-rater reliability (Cohen's Kappa)

Programming & Tools: Python, PyTorch, R, MATLAB, SQL, PySpark, AWS (EC2, S3, Athena), Google Cloud Platform (Gemini API, Cloud TTS), Databricks, MLflow, Weights & Biases, Docker, FastAPI, scikit-learn, Tableau